Relational Algebra Problems

Problem 1: Library

Schemas:

- Book(BookID, Title, Author)
- Borrower(BorrowerID, Name)
- Loan(BorrowerID, BookID)

Q: Find the names of all borrowers who have borrowed at least one book.
Solution:

$$\begin{aligned} Borrowed &:= \pi_{\text{BorrowerID}}(\text{Loan}) \\ Result &:= \pi_{\text{Name}}(\text{Borrower} \bowtie \text{Borrowed}) \end{aligned}$$

Q: Find the names of borrowers who have borrowed books authored by 'George Orwell'.
Solution:

$$\begin{aligned} OrwellBooks &:= \pi_{\text{BookID}}(\sigma_{\text{Author}='GeorgeOrwell'}(\text{Book})) \\ OrwellLoans &:= \text{Loan} \bowtie OrwellBooks \\ Result &:= \pi_{\text{Name}}(\text{Borrower} \bowtie OrwellLoans) \end{aligned}$$

Problem 2: University

Schemas:

- Student(StudentID, Name, Major)
- Course(CourseID, Title)
- Enrolled(StudentID, CourseID)

Q: Find the names of students majoring in 'Computer Science'.
Solution:

$$Result := \pi_{\text{Name}}(\sigma_{\text{Major}='ComputerScience'}(\text{Student}))$$

Q: Find the names of students enrolled in the course titled 'Database Systems'.
Solution:

$$\begin{aligned} DBCourse &:= \pi_{\text{CourseID}}(\sigma_{\text{Title}='DatabaseSystems'}(\text{Course})) \\ EnrolledInDB &:= \text{Enrolled} \bowtie DBCourse \\ Result &:= \pi_{\text{Name}}(\text{Student} \bowtie EnrolledInDB) \end{aligned}$$

Problem 3: Airline

Schemas:

- Passenger(PassengerID, Name)
- Flight(FlightID, Destination)
- Booking(PassengerID, FlightID)

Q: Find names of passengers who booked a flight to 'Paris'.
Solution:

$$\begin{aligned} ParisFlights &:= \pi_{\text{FlightID}}(\sigma_{\text{Destination}='Paris'}(\text{Flight})) \\ ParisBookings &:= \text{Booking} \bowtie \text{ParisFlights} \\ Result &:= \pi_{\text{Name}}(\text{Passenger} \bowtie \text{ParisBookings}) \end{aligned}$$

Q: Find the names of passengers who have booked a flight to 'Tokyo' or 'Sydney'.
Solution:

$$\begin{aligned} DestFlights &:= \pi_{\text{FlightID}}(\sigma_{\text{Destination}='Tokyo' \vee \text{Destination}='Sydney'}(\text{Flight})) \\ BookedFlights &:= \text{Booking} \bowtie DestFlights \\ Result &:= \pi_{\text{Name}}(\text{Passenger} \bowtie BookedFlights) \end{aligned}$$

Problem 4: Company

Schemas:

- Employee(EmpID, Name, Salary)
- Department(DeptID, DeptName)
- WorksIn(EmpID, DeptID)

Q: Find names of employees who earn more than 50000.
Solution:

$$Result := \pi_{\text{Name}}(\sigma_{\text{Salary}>50000}(\text{Employee}))$$

Q: Find names of employees who work on projects outside their own department.
Solution:

$$\begin{aligned} EmployeeProject &:= \text{Employee} \bowtie \text{WorksOn} \bowtie \text{Project} \\ OutsideDept &:= \sigma_{\text{Employee.DeptID} \neq \text{Project.DeptID}}(EmployeeProject) \\ Result &:= \pi_{\text{Name}}(OutsideDept) \end{aligned}$$

Problem 5: Music Streaming

Schemas:

- Listener(ListenerID, Name)
- Song(SongID, Title, Artist)
- Played(ListenerID, SongID)

Q: Find names of listeners who have played a song by 'Taylor Swift'.
Solution:

$$\begin{aligned} TaylorSongs &:= \pi_{\text{SongID}}(\sigma_{\text{Artist}='TaylorSwift'}(\text{Song})) \\ Plays &:= \text{Played} \bowtie \text{TaylorSongs} \\ Result &:= \pi_{\text{Name}}(\text{Listener} \bowtie \text{Plays}) \end{aligned}$$

Q: Find names of listeners who have listened to songs by both 'Adele' and 'Ed Sheeran'.
Solution:

$$\begin{aligned} AdeleSongs &:= \pi_{\text{SongID}}(\sigma_{\text{Artist}='Adele'}(\text{Song})) \\ EdSongs &:= \pi_{\text{SongID}}(\sigma_{\text{Artist}='EdSheeran'}(\text{Song})) \\ AdeleListeners &:= \pi_{\text{ListenerID}}(\text{Played} \bowtie AdeleSongs) \\ EdListeners &:= \pi_{\text{ListenerID}}(\text{Played} \bowtie EdSongs) \\ BothListeners &:= AdeleListeners \cap EdListeners \\ Result &:= \pi_{\text{Name}}(\text{Listener} \bowtie BothListeners) \end{aligned}$$